

Internal technical plan

A Nontrivial Assay Program for Type-1, $N = 3$ Landau–Zener Continuation-Functor Constructors and Simple Component Structure

This note records the corrected test standard after the previous 520-example segmented-product assay was recognized to be tautological as a validation of local constructors. It specifies what a nontrivial assay must do, separates three increasingly independent constructor levels, lists what has already been validated, what has not been validated, and what has been falsified, and gives a maximalist parallel-agent dispatch plan.

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1 Executive summary

The current state is best summarized by the distinction

$$\boxed{\text{segmented exact propagator plumbing}} \neq \boxed{\text{independent continuation-functor constructor.}}$$

The earlier 520-example assay confirmed that, once geometric window centers and bounds are chosen, exact segment matrices multiply back to the one-pass fundamental matrix at roundoff-level precision. That test was valuable for debugging factor order, endpoint permutations, and bookkeeping. It did *not* validate any independently constructed local Q_4 window object, because the factors were obtained from the same global fundamental solution.

A nontrivial assay must therefore satisfy two conditions.

1. The benchmark one-pass integration and the constructor used in the product must be disjoint computational pathways.
2. The product factors must be shown to have simple structure, not merely to be arbitrary full 3×3 matrices solved on small intervals.

The corrected no-active-selector scattering ansatz is

$$S_{\text{noSel}}^{\text{ad}} = \Pi_+ D_2 \mathcal{W}_{X_2} D_1 \mathcal{W}_{X_1} D_0 \Pi_-, \quad (1)$$

where X_1, X_2 are the two projected conjugate-pair Q_4 windows in real-axis order. The exact real crossing $L_H = 0$ is a regular marker and is absorbed into whichever edge contains it. The old point-local replacement of each $\mathcal{W}_X$ by one embedded $2 \oplus 1$ Airy block is falsified because it produces structural zero amplitudes not present in the ODE data.

The three nontrivial assay levels are:

- Level 1.** Independently solved segment ODEs. This avoids using the stored one-pass fundamental matrix to build factors, but it still uses the same differential equation locally and is therefore a numerical-constructor assay rather than a structural analytic assay.
- Level 2.** Lateral/median full-window constructors. Each projected conjugate Q_4 window is built as a median of upper and lower lateral continuations, $\mathcal{W}_X = \text{Med}(U_X^+, U_X^-)$, computed independently from lateral contour IVPs.
- Level 3.** Analytic or semi-analytic Q_4 window constructors. Each $U_X^\pm$ is replaced by local normal-form data, Abelian phase integrals, and independently constructed entry/exit frame maps. This is the first level that would provide a genuine closed or semi-closed solution mechanism.

2 Mathematical baseline

The Type-1, $N = 3$ problem is controlled by the rational spectral data

$$u(\lambda) = \frac{n(\lambda)}{p(\lambda)}, \quad E(\lambda) = \frac{m(\lambda)}{p(\lambda)}, \quad q_u(\lambda) = up(\lambda) - n(\lambda). \quad (2)$$

The gauged adiabatic amplitudes solve

$$f'_i(u) = - \sum_{j \neq i} S_{ij}(u) f_j(u) + i(E_i(u) - E_x(u)) f_i(u). \quad (3)$$

The primary benchmark observable is the transition matrix

$$P_{x \rightarrow j} = |f_j(+\infty)|^2. \quad (4)$$

The quotient-ring formulation packages the reduced carrier in

$$\mathcal{E}_u = \mathbb{C}[\lambda]/(q_u(\lambda)), \quad \partial_u[R] = i[(m - E_x p)s_u R], \quad ps_u \equiv 1 \pmod{q_u}. \quad (5)$$

The same geometry is organized by the commuting multiplication operators

$$Y = M_{Q_4 L_H^2/p^2}, \quad R = M_{W_4/L_H^4}. \quad (6)$$

Thus Q_4 controls the phase-square-root sector, W_4 controls transport ramification, and L_H is a member-dependent phase/crossing factor. The common/spinor cover carries

$$\omega = \sqrt{y} du, \quad \eta = S du = \frac{\rho}{y} du. \quad (7)$$

Away from the transport discriminant, the spectral projectors of R give canonical gluing of the pair sectors. Active selector points are a different mechanism: $W_4 = 0$ plus phase resonance.

3 Corrected product architecture

3.1 No-active-selector chamber

For real generic parameters and $\theta = 0$, the four Q_4 roots typically occur as two conjugate pairs. Each pair defines one projected real-axis window. Let

$$X_1 \prec X_2$$

be the order of those two windows under the projection coordinate

$$\tau(z; \theta) = \Re(e^{-i\theta} z). \quad (8)$$

For window X_k choose a real projected interval

$$I_{X_k} = [\ell_{X_k}, r_{X_k}], \quad (9)$$

with disjoint interiors. Then the no-selector product is Eq. (1), with

$$D_0 : \quad -\infty \rightarrow \ell_{X_1}, \quad (10)$$

$$D_1 : \quad r_{X_1} \rightarrow \ell_{X_2}, \quad (11)$$

$$D_2 : \quad r_{X_2} \rightarrow +\infty. \quad (12)$$

The exact crossing $L_H = 0$ is not a factor. It remains a useful marker for stationary-phase and large-slope asymptotics, but it is absorbed into the edge containing $u_c = u(\lambda_c)$.

3.2 General active-selector chamber

If active selectors are present, the product becomes

$$S^{\text{ad}} = \Pi_+ D_m V_m D_{m-1} \cdots V_1 D_0 \Pi_-, \quad (13)$$

where each V_k is one of:

1. a full projected Q_4 window $\mathcal{W}_X$;
2. an embedded scalar active-selector block $J_{v,\theta}(q; H)$;
3. a conventional ordinary turning block if the chosen formulation resolves individual ordinary turning points rather than projected windows.

In a generic simple active selector case,

$$J_{v,\theta}(q; H) = \iota_v(I + s_{v,\theta}^{\text{BS}}(q; H)N_{v,\theta}), \quad (14)$$

with all regular sector propagation assigned to adjacent D -factors, not to the pure selector block.

4 What previous computations actually established

Claim	Status	Reason
Bookkeeping segmentation and path order are implementable.	Validated at Level 0	520 samples reproduced one-pass probabilities after segment factors were reconstructed from the global one-pass matrix. This validates indexing, boundaries, multiplication order, and endpoint permutations.

Generic selectors are inactive at $\theta = 0$ away from walls.	Empirically supported	In the 520-sample generic assay, no selectors were active at tolerance 10^{-8} . This matches the expectation that phase activation is codimension one for fixed θ .
The exact $L_H = 0$ crossing is a point jump.	Falsified	The crossing shrink diagnostic shows the finite crossing-window propagator tends to identity as $h_c \rightarrow 0$ at fixed H .
A product containing only two embedded point-local $2 \oplus 1$ Q_4 Airy blocks is sufficient.	Falsified	That algebraic collapse forces structural zeros. The ODE transition matrices in the logged samples show no such zero entries.
Each projected Q_4 window is a full continuation object.	Required, not yet analytically constructed	It is necessary to remove structural zeros, but its independent analytic or semi-analytic construction remains the target of Levels 2–3.
Active selector insertion is a pure embedded scalar block in generic simple cases.	Symbolically established in separate selector notes	The local-to-global theorem applies to active selectors, not to ordinary projected Q_4 windows.

5 Control level: exact segmentation identity

Before the three nontrivial assay levels, retain a Level 0 control. Let $U(t, -T)$ be the benchmark one-pass fundamental matrix. For any boundaries

$$-T = t_0 < t_1 < \cdots < t_m = T,$$

construct segments by

$$F[t_b, t_a] = U(t_b, -T)U(t_a, -T)^{-1}. \quad (15)$$

Then

$$U(T, -T) = F[t_m, t_{m-1}] \cdots F[t_1, t_0]. \quad (16)$$

This is mathematically tautological. It should remain in the harness as a smoke test, but no claim of independent continuation construction should be based on it.

6 Level 1: independently solved segment-ODE assay

6.1 Purpose

Level 1 enforces the minimum separation between benchmark and constructor. It tests whether independently solved edge and window IVPs compose correctly without reusing stored values from the

benchmark fundamental matrix.

6.2 Inputs

For each sample:

- real ordered $\epsilon_0 < \epsilon_1 < \epsilon_2$;
- nonzero real γ_i ;
- real slopes a_i ;
- Borel direction θ , initially $\theta = 0$;
- projected Q_4 windows $I_X = [\ell_X, r_X]$;
- active-selector windows if present.

6.3 Constructor rules

The benchmark solves Eq. (3) once over $[-T, T]$ for each incoming channel. The constructor must not read benchmark intermediate values. For each segment $[a, b]$, independently solve

$$\frac{dU_{[a,b]}}{du} = \mathcal{A}(u)U_{[a,b]}, \quad U_{[a,b]}(a) = I, \quad (17)$$

where $\mathcal{A}$ is the chosen gauged or ungauged coefficient matrix. The product is formed from these independently initialized segment matrices.

6.4 Pass criteria

1. For every sample and every transition probability,

$$\max_{x,j} |P_{x \rightarrow j}^{\text{prod}} - P_{x \rightarrow j}^{\text{direct}}| < 10^{-8}.$$

2. The product must remain stable under shrinking or expanding windows within admissible non-overlap ranges.
3. The result must remain stable under refining the cutoff T and solver precision.

6.5 What Level 1 proves and does not prove

Level 1 proves that the factorization is not merely the algebraic identity $U_b U_a^{-1}$. It still does not prove simple component structure. Full-window matrices are still arbitrary local ODE propagators unless further reduced.

7 Level 2: lateral/median full-window assay

7.1 Purpose

Level 2 tests the actual real-axis continuation idea. A projected Q_4 conjugate-pair window should be reconstructed as a median of independent upper and lower lateral continuations:

$$\mathcal{W}_X = \text{Med}(U_X^+, U_X^-) = U_X^+ ((U_X^-)^{-1} U_X^+)^{-1/2}. \quad (18)$$

The matrix square root uses a fixed branch, with branch-change events logged as chamber changes.

7.2 Constructor rules

For each projected Q_4 window $X = \{q, \bar{q}\}$:

1. Choose common real projected endpoints ℓ_X, r_X .
2. Construct two contours γ_X^+ and γ_X^- from ℓ_X to r_X , passing through the upper and lower lateral sectors of $u(q)$ and $u(\bar{q})$ respectively.
3. Solve the same local ODE independently on those contours with identity initial matrix:

$$U_X^\pm = \mathcal{P} \exp \int_{\gamma_X^\pm} \mathcal{A}(u) du.$$

4. Form $\mathcal{W}_X$ by Eq. (18).

Edge factors may be either independent real-edge ODE solves or Abelian-frame edge factors with independently constructed endpoint frame maps. The direct benchmark is still the one-pass real-axis ODE.

7.3 Simplicity tests at Level 2

A successful Level 2 should show more than accuracy. For each $\mathcal{W}_X$ log:

- conjugation symmetry: $U_X^- \approx \overline{U_X^+}$ in the compatible real frame;
- median stability under contour deformation within the same chamber;
- low-dimensional variation of $\log \mathcal{W}_X$ under changes of window width and contour curvature;
- absence of forced structural zeros;
- bounded condition number of the Stokes automorphism $(U_X^-)^{-1} U_X^+$;
- agreement of the real-axis product with benchmark probabilities to 10^{-8} or better.

7.4 Failure modes

Level 2 fails if median branch choices jump unpredictably, if lateral contours do not yield a stable median, or if the product requires arbitrary full 3×3 fitting beyond independent lateral IVPs. A failure here does not falsify the Type-1 geometry; it falsifies this median implementation.

8 Level 3: analytic or semi-analytic Q_4 window assay

8.1 Purpose

Level 3 is the first genuinely structural assay. It tries to replace $U_X^\pm$ by simple building blocks:

$$U_X^\pm = F_{X,\text{out}}^\pm C_X^\pm P_X^\pm (F_{X,\text{in}}^\pm)^{-1}. \quad (19)$$

Here:

- $C_X^\pm$ is a universal ordinary Airy/Stokes connection factor in a local active-pair normal form;
- $P_X^\pm$ is diagonal or block-diagonal Abelian propagation built from integrals of ω and possible half-form/transport terms;
- $F_{X,\text{in/out}}^\pm$ are entry/exit frame maps from the physical adiabatic or quotient-ring frame to the local sectorial normal-form frame.

8.2 Allowed information

The constructor may use:

- algebraic data $p, n, m, Q_4, W_4, L_H, \kappa_S$;
- roots of Q_4 and W_4 ;
- Abelian integrals on $\hat{\mathcal{C}}$;
- local Airy/Langer normal forms at Q_4 roots;
- canonical gluing projectors from $\mathbf{R}$;
- local frame equations solved only in small overlap annuli, not across the whole projected window.

The constructor must not use the one-pass benchmark fundamental matrix or exact window IVPs except as validation references.

8.3 Simplicity standard

Level 3 is successful only if each $\mathcal{W}_X$ is described by a small list of interpretable data:

$$\mathcal{W}_X = \text{Med}(F_{\text{out}}^+ C_X^+ P_X^+ (F_{\text{in}}^+)^{-1}, F_{\text{out}}^- C_X^- P_X^- (F_{\text{in}}^-)^{-1}), \quad (20)$$

with $C_X^\pm$ universal or near-universal, $P_X^\pm$ Abelian, and F -maps algebraic or short local-frame solves. An arbitrary numerically fitted full 3×3 matrix does not pass.

8.4 Pass criteria

1. Accuracy: all transition probabilities below 10^{-8} error against one-pass ODE on more than 500 generic samples and targeted stress samples.
2. Simplicity: each factor admits the claimed decomposition with logged scalar or low-dimensional parameters.
3. Stability: changing local radii and contour gauges conjugates adjacent F maps but leaves the product invariant.
4. Falsifiability: every failure must be logged as one of a finite list – median branch failure, frame-map failure, Airy normal-form failure, edge phase failure, selector activation error, endpoint permutation error, or true counterexample.

9 Dense parameter sampling

A dense assay should combine random and adversarial sampling. Recommended strata:

1. Generic real ordered ϵ with γ_i away from zero and moderate slopes.
2. Small-slope and large-slope regimes.
3. Near-degenerate but accepted Q_4 roots, excluding collisions only by a controlled margin.
4. Near W_4 phase-resonance walls, both active and inactive selectors.
5. Exact crossing u_c placed before, between, and after the two projected Q_4 windows.
6. Crossing near $u = 0$ and near projected Q_4 windows.
7. High-curvature transport ramification points.
8. Endpoint/asymptotic stress with increasing T and asymptotic phase normalization checks.

A run should record accepted samples and rejected samples separately. Rejection reasons must be algebraic, not outcome-based: repeated roots, near collisions beyond precision budget, path crossing ambiguity, singular contour, or failed precision certificate.

10 Mandatory logs and schemas

Every run should emit:

- parameter vectors (ϵ, γ, a) ;
- Q_4 , W_4 , and L_H roots and projected u values;
- selector residuals and active selector list;
- window bounds ℓ_X, r_X and edge bounds;
- all endpoint permutations $\Pi_{\pm}$;
- direct benchmark transition matrix;
- product transition matrix;
- per-entry absolute errors and row/column unitarity defects;
- constructor-specific diagnostics: condition numbers, median eigenvalues, branch choices, contour hashes, frame-map gauges, and simplicity parameters.

For Level 3, the log must also contain the factorization residual

$$\|U_X^{\pm} - F_{\text{out}}^{\pm} C_X^{\pm} P_X^{\pm} (F_{\text{in}}^{\pm})^{-1}\| \quad (21)$$

computed against an independently solved local-contour IVP, never against the one-pass benchmark.

11 Maximalist parallel-agent dispatch plan

The maximalist assay can be dispatched as a federation of autonomous agents. Each agent writes machine-readable artifacts and is forbidden to consume artifacts that would create circular validation.

Agent	Task	Outputs and independence rule
Parameter factory	Generate generic and adversarial parameter batches.	CSV/JSON parameter batches plus algebraic rejection reasons. No access to ODE outputs.
Geometry auditor	Compute p, n, m, Q_4, W_4, L_H , roots, projections, degeneracy scores.	Geometry logs and certified window order. No access to transition matrices.
Benchmark ODE agent	Compute one-pass high-precision transition matrices.	Direct P matrices, unitarity defects, solver logs. No access to product factors.

Level 1 constructor agents	Solve independent real-edge and full-window segment IVPs with identity initial data.	Segment matrices and product errors. No use of stored benchmark intermediates.
Lateral contour agents	For each Q_4 window, compute U_X^+ and U_X^- on independent contours.	Lateral matrices, contour certificates, median factors. No access to benchmark.
Median auditor	Check matrix square-root branches, conjugation symmetry, contour deformation invariance.	Median diagnostics, branch logs, failure classifications.
Frame-map agents	Construct $F_{\text{in/out}}^\pm$ using quotient-ring frames, projectors, or local gauge equations.	Frame maps and local residuals. No direct use of full-window exact IVP except validation reference.
Airy normal-form agents	Compute local Airy/Langer data at Q_4 points and selector points.	$C_X^\pm$, local actions, Stokes constants, normal-form errors.
Edge Abelian agents	Compute edge propagation from Abelian integrals and endpoint frame maps.	D_e factors, asymptotic residue/finite-part logs.
Selector agents	Detect active W_4 selectors and insert scalar blocks when active.	Selector residuals, local multipliers, insertion labels.
Crossing audit agents	Verify $L_H = 0$ shrink-to-identity across random and targeted samples.	$B_c(h)$ shrink ratios and marker placement.
Falsification agents	Search for structural zeros, instability, branch flips, near-collision failures.	Counterexample candidates with minimal reproducing data.
Reconciliation agent	Compose product factors and compare to benchmark.	Product matrices and all per-entry errors. Cannot construct factors.
Archive agent	Bundle logs, code, seeds, and provenance.	Reproducible audit bundle. No mathematical decisions.

12 Acceptance ladder

The assay should be graded on the following ladder.

Bookkeeping success. Level 0 reproduces the benchmark. Already achieved, but tautological.

Independent numerical success. Level 1 reproduces benchmark with independent segment IVPs.

Continuation success. Level 2 reproduces benchmark using lateral/median full-window objects.

Structural success. Level 3 reproduces benchmark using simple local normal-form, Abelian, and frame-map components.

Theorem-shaping success. All failures are localized to nongeneric collision regimes or to explicitly named missing subconstructors.

13 Conclusion

The nontrivial assay should not ask whether exact propagators can be split and multiplied. That is guaranteed. It should ask whether the Type-1, $N = 3$ geometry supplies independently constructible and structurally simple factors for the scattering matrix. The corrected program is therefore:

$$\boxed{\text{one-pass ODE benchmark}} \quad \text{versus} \quad \boxed{\Pi_+ D_2 \mathcal{W}_{X_2} D_1 \mathcal{W}_{X_1} D_0 \Pi_-}$$

with $\mathcal{W}_X$ built first by independent lateral/median continuation and then, at the decisive level, by analytic or semi-analytic normal-form data.

A Pseudo-code skeleton

```

for sample in parameter_batch:
    geom = GeometryAgent(sample)
    if geom.reject: log_reject(); continue

    P_direct = BenchmarkODE(sample, geom, T, precision)

    # Level 1
    factors_L1 = IndependentSegmentODE(sample, geom.windows, geom.edges)
    P_L1 = ComposeAndPermute(factors_L1, geom.Pi_plus, geom.Pi_minus)
    compare(P_L1, P_direct)

    # Level 2
    lateral = LateralContourConstructors(sample, geom.Q4_windows)
    W = {X: Med(lateral[X].U_plus, lateral[X].U_minus) for X in windows}
    edges = IndependentEdgeConstructors(sample, geom.edges)
    P_L2 = ComposeAndPermute(edges, W, geom.Pi_plus, geom.Pi_minus)
    compare(P_L2, P_direct)

    # Level 3
    normal_forms = AiryAndFrameConstructors(sample, geom.Q4_windows)
    W_analytic = BuildWindowMedians(normal_forms)
    edges_abelian = AbelianEdgeConstructors(sample, geom.edges)
    P_L3 = ComposeAndPermute(edges_abelian, W_analytic, geom.Pi_plus, geom.Pi_minus)
    compare(P_L3, P_direct)

```

B Endpoint permutation convention

The convention used in the previous executable scaffold was

$$\Pi_- = I_3, \quad \Pi_+ = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (22)$$

The nontrivial assay should keep this convention explicit and should also run a permutation audit that independently confirms the incoming and outgoing asymptotic label maps for every sample class.